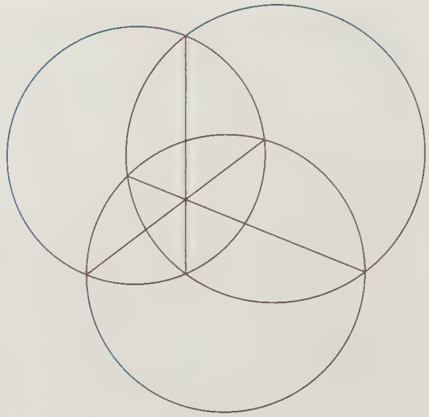


Radical Axis Theorem

Power corrupts. Absolute power is kind of neat. – John Lehman, Secretary of the Navy, 1981-1987

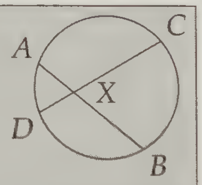
CHAPTER 13 Power of a Point

In the previous chapter we explored the angles formed by lines that meet inside, outside, or on a circle. In this chapter, we explore the lengths of segments that meet inside or outside a circle. Together, the relationships we will prove and use in this chapter are given the lofty name 'Power of a Point.'

13.1 What is Power of a Point?

Problems

Problem 13.1: Chords $\overline{AB}$ and $\overline{CD}$ meet at point X as shown. In this problem we will prove that



$$(XA)(XB) = (XC)(XD).$$

(a) Draw $\overline{AD}$ and $\overline{BC}$. Why is $\angle XDA = \angle XBC$?

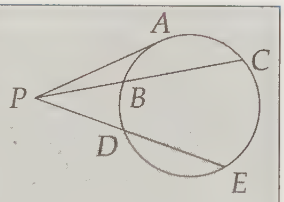
(b) Show that the two triangles in the diagram are similar. Use these similar triangles to prove that $XA/XC = XD/XB$.

(c) Complete the proof that $(XA)(XB) = (XC)(XD)$.

(d) Is every chord of the circle that passes through X split into two pieces that have a product equal to $(XA)(XB)$? (In other words, is there anything special about chord $\overline{CD}$ or will this proof work for every chord through X?)

Problem 13.2: Point P is outside a given circle and A is on the circle such that $\overline{PA}$ is tangent to the circle. C and E are on the circle such that secants $\overline{PC}$ and $\overline{PE}$ intersect the circle again at B and D, respectively. In this problem we prove that

$$PA^2 = (PB)(PC) = (PD)(PE).$$



(a) Draw $\overline{AB}$ and $\overline{AC}$. Why does $\angle ACP = \angle BAP$?

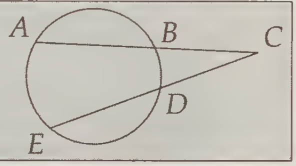
(b) Use the angle equality from the first part to identify two triangles that are similar.

(c) Use the triangle similarity to prove $PA^2 = (PB)(PC)$.

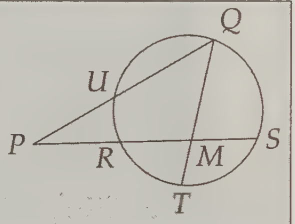
(d) How can we show that $(PD)(PE) = (PB)(PC)$?

Problem 13.3: Chords $\overline{TY}$ and $\overline{OP}$ meet at point K such that $TK = 2$, $KY = 16$, and $KP = 2(KO)$. Find OP.

Problem 13.4: In the diagram, $CB = 9$, $BA = 11$, and $CE = 18$. Find DE.



Problem 13.5: Points R and M trisect $\overline{PS}$, so $PR = RM = MS$. Point U is the midpoint of $\overline{PQ}$, $TM = 2$, and $MQ = 8$.



(a) Find RM and MS.

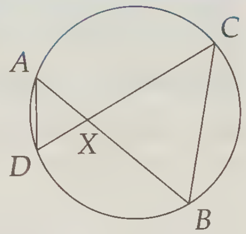
(b) Use the relationship you found in Problem 13.2 to find PU. (Don't forget that U is the midpoint of $\overline{PQ}$.)

We begin this section by proving the Power of a Point Theorem in two parts. First, we consider points inside a circle.

Problem 13.1: Given chords $\overline{AB}$ and $\overline{CD}$ that meet at point X, prove that

$$(XA)(XB) = (XC)(XD).$$

Solution for Problem 13.1: When we rearrange what we want to prove, we have $XA/XC = XD/XB$. The ratios suggest we look for similar triangles, so we draw $\overline{AD}$ and $\overline{BC}$ to form triangles. Since $\angle B$ and $\angle D$ are inscribed in the same arc, we have $\angle B = \angle D$. We also have $\angle AXD = \angle CXB$, so we have $\triangle AXD \sim \triangle CXB$ by AA Similarity. Therefore, we have $XA/XC = XD/XB$, so $(XA)(XB) = (XC)(XD)$.

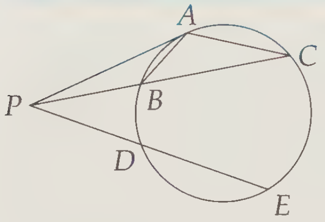


Note that there's nothing special about chords $\overline{AB}$ and $\overline{CD}$. We can use the same exact approach as above to show that each of this circle's chords through X is divided into two pieces such that the product of the lengths of these pieces equals $(XA)(XB)$. □

Having finished with points inside the circle, we turn to points outside the circle.

Problem 13.2: Point P is outside a given circle and A is on the circle such that $\overline{PA}$ is tangent to the circle. C and E are on the circle such that secants $\overline{PC}$ and $\overline{PE}$ intersect the circle again at B and D, respectively. Prove that

$$PA^2 = (PB)(PC) = (PD)(PE).$$



Solution for Problem 13.2: We start as we did in the previous problem, drawing $\overline{AB}$ and $\overline{AC}$ to make triangles. Since $\angle ACB = \overparen{AB}/2$ and $\angle PAB = \overparen{AB}/2$, we have $\triangle PAB \sim \triangle PCA$ by AA Similarity (since $\angle P$ is the same in both triangles as well). Therefore, $PA/PB = PC/PA$, so $PA^2 = (PB)(PC)$.

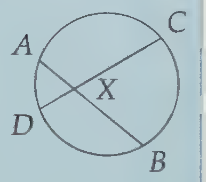
Just as there was nothing special about the chords we looked at in the previous problem, there's nothing special about secant $\overline{PC}$ here. By exactly the same reasoning, we have $(PD)(PE) = PA^2$, so $(PD)(PE) = (PB)(PC)$. Notice that once again there's nothing special about the tangent and the secants we have chosen except that they have point P in common. □

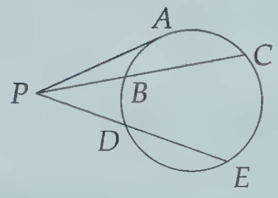
We can put the facts we've proved in the last two problems together in a single statement.

Important: Suppose a line through a point P intersects a circle in two points, U and V. The **Power of a Point Theorem** states that for all such lines, the product $(PU)(PV)$ is constant. We call this product the **power** of point P.

For example, in the figure at right, applying Power of a Point to X with respect to the circle shown gives

$$(XA)(XB) = (XC)(XD).$$





In the figure at left, the power of point P with respect to the circle gives us

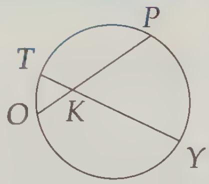
$$PA^2 = (PB)(PC) = (PD)(PE).$$

We think of Power of a Point whenever we have a problem involving intersecting segments and a circle.

Let's try using Power of a Point on a few problems.

Problem 13.3: Chords $\overline{TY}$ and $\overline{OP}$ meet at point K such that $TK = 2$, $KY = 16$, and $KP = 2(KO)$. Find OP.

Solution for Problem 13.3: A quick sketch suggests how to apply Power of a Point. From the power of point K, we have



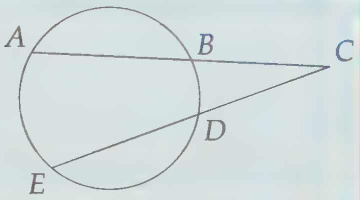
$$(KP)(KO) = (KT)(KY).$$

Substituting the given information in this equation yields

$$2(KO)(KO) = 2(16),$$

from which we find $KO = 4$. Therefore, $KP = 2KO = 8$, so $OP = KO + KP = 12$. □

Problem 13.4: In the diagram, $CB = 9$, $BA = 11$, and $CE = 18$. Find DE.

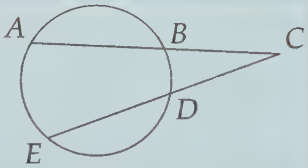


Solution for Problem 13.4: We have two intersecting secants, so we apply Power of a Point, which gives

$$(CD)(CE) = (CB)(CA).$$

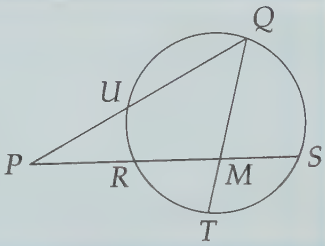
Therefore, $(CD)(18) = 9(20)$, so $CD = 10$ and $DE = CE - CD = 8$. □

WARNING!! A very common mistake in applying Power of a Point is to write $(BC)(BA) = (DC)(DE)$ when faced with the figure at right. This alleged equality is **not** what Power of a Point tells us! Note that whenever we write a Power of a Point relationship, the same point must appear in *all* of the segments in the equation, as point C does when we write the correct relationship for the secants in the figure above:



$$(CD)(CE) = (CB)(CA).$$

Problem 13.5: Points R and M trisect $\overline{PS}$, so $PR = RM = MS$. Point U is the midpoint of $\overline{PQ}$, $TM = 2$, and $MQ = 8$. Find PU.



Solution for Problem 13.5: Circles, chord lengths, and secant lengths. This is a job for Power of a Point. The power of point M gives us

$$(MR)(MS) = (MT)(MQ).$$

We know that $RM = MS$, so substitution gives $MR^2 = (2)(8)$, i.e., $MR = 4$. Therefore, $PR = MR = 4$ and $PS = 3(MR) = 12$. Since U is the midpoint of $\overline{PQ}$, we have $PQ = 2PU$. Now we can apply the power of point P to find:

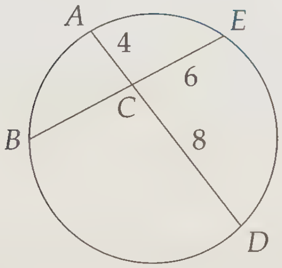
$$(PU)(PQ) = (PR)(PS).$$

Substitution gives $(PU)(2PU) = 4(12)$, so $PU = 2\sqrt{6}$. □

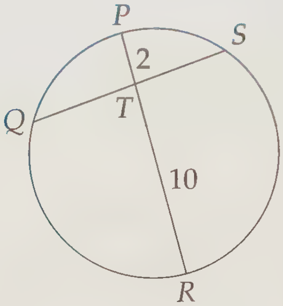
Extra! *Never say of a branch of mathematics, 'There's something I don't need to know.' It always comes back to haunt you.*
–Charles Rickart

Exercises

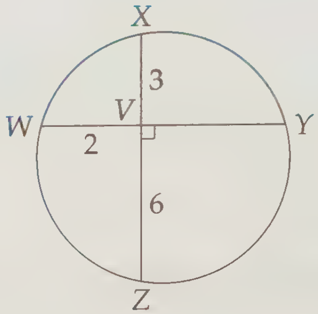
13.1.1



(a) Find BC.

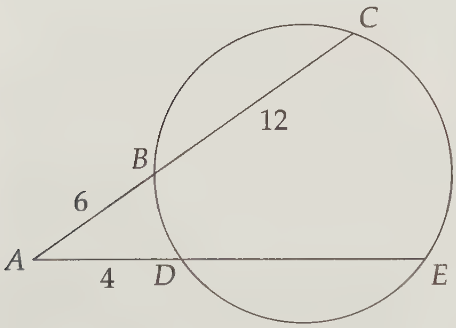


(b) $QS = 9$ and $TQ < TS$; find TQ.

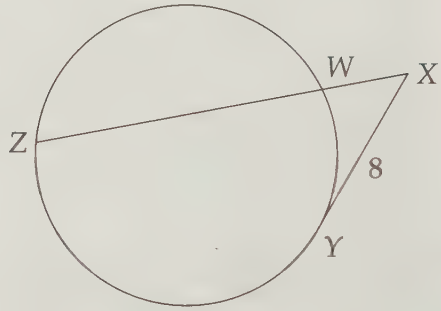


(c) Find ZY.

13.1.2



(a) Find DE.

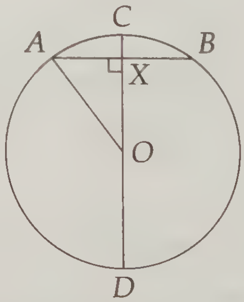


(b) Given $XZ = 12$, find WZ.

13.1.3 Chords $\overline{XY}$ and $\overline{AB}$ intersect at Q. Given $AQ = 4$, $BQ = 6$, and $XQ = 8$, find XY.

13.1.4 Chord $\overline{UV}$ bisects chord $\overline{ST}$ at point M. Given $ST = 12$ and $UV = 15$, find all possible values of UM.

13.1.5 In this problem we use Power of a Point to prove the Pythagorean Theorem.

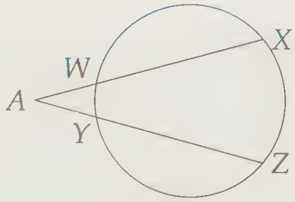


(a) Chord $\overline{AB}$ is perpendicular at point X to diameter $\overline{CD}$ of circle O. Let the radius of the circle be c, the length of $\overline{AB}$ be $2a$, and $OX = b$. Find CX and XD in terms of b and c.

(b) Use Power of a Point to show that $c^2 = a^2 + b^2$.

(c) How can you use this argument to prove the Pythagorean Theorem? **Hints:** 3, 41

13.1.6★ In the figure at right, we have $WX = YZ$. Prove that $AW = AY$. **Hints:** 62

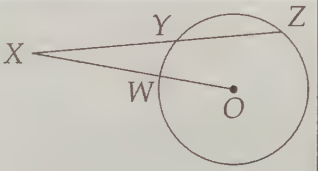


13.2 Power of a Point Problems

We now apply Power of a Point to some more challenging problems. As we'll see, Power of a Point can be particularly useful for proofs involving segment lengths when circles are part of the problem.

Problems

Problem 13.6: Given that $XY = 6$, $YZ = 5$, and that point X is 9 units from the center, O, of the circle, we wish to find the area of the circle.



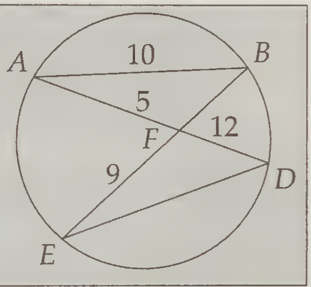
(a) Extend $\overline{XO}$ to meet the circle again at V.

(b) Let the radius of the circle be r; use the power of point X to find r^2, and hence, the area of the circle.

Problem 13.7:

(a) Find BF.

(b) Find ED.

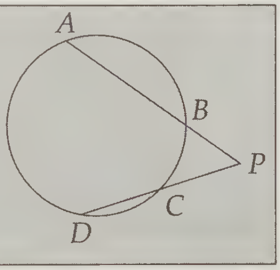


Problem 13.8: In the diagram, we have $BP = 8$, $AB = 10$, $CD = 7$, and $\angle APC = 60°$. *(Source: AHSME)*

(a) Find CP.

(b) Show that $\triangle ACP$ is a 30-60-90 triangle. **Hints:** 317

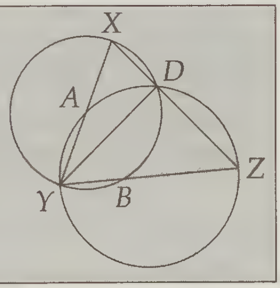
(c) Find the area of the circle.



Problem 13.9: In this problem we wish to show that $XA = BZ$ in the diagram at right, given that $\overline{DY}$ bisects $\angle XYZ$.

(a) Use Power of a Point to find expressions for XA and BZ in terms of other segments in the diagram.

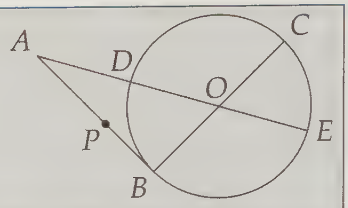
(b) Use the Angle Bisector Theorem to prove that your expressions from the first part are equal.



Problem 13.10: Point O is the center of the circle, $\overline{AB} \perp \overline{BC}$, $AP = AD$, and $\overline{AB}$ has length twice the radius of the circle. Prove that

$$AP^2 = (PB)(AB).$$

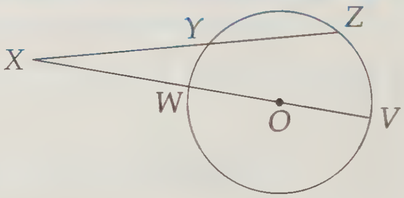
(*Source: AHSME*)



We start where we left off last section by finding lengths with Power of a Point in slightly more complicated problems.

Problem 13.6: Z is on $\odot O$ and secant $\overline{XZ}$ hits the circle at Y such that $XY = 6$ and $YZ = 5$. Given that X is 9 units from the center of this circle, find the area of the circle.

Solution for Problem 13.6: We start with a diagram. We continue segment $\overline{XO}$ to hit the circle again at V, so that we can use the power of point X: $(XW)(XV) = (XY)(XZ)$. If we let the radius of the circle be r, substitution gives $(XO-r)(XO+r) = 6(6+5)$. Since $XO = 9$, we have $(9-r)(9+r) = 66$, from which we find $r^2 = 15$. Therefore, the area of the circle is $\pi r^2 = 15\pi$. □

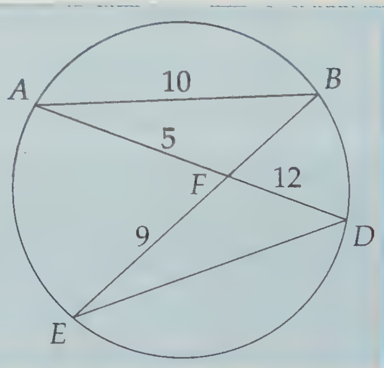


Concept: If a problem includes a segment that abruptly stops in the middle of a circle, consider continuing the segment until it hits the circle. Then see what Power of a Point gives you.

Problem 13.7:

(a) Find BF.

(b) Find ED.



Solution for Problem 13.7:

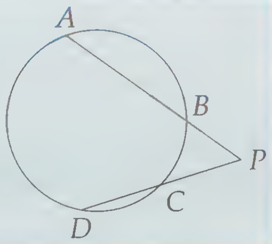
(a) We can find BF with Power of a Point: $(FE)(FB) = (FA)(FD)$, so $FB = (FA)(FD)/(FE) = 20/3$.

(b) While Power of a Point won't get us anywhere with ED, our proof of Power of a Point guides us to the answer. Specifically, since $\angle FDE = \angle FBA$ and $\angle AFB = \angle EFD$, we have $\triangle AFB \sim \triangle EFD$, so $ED/BA = EF/AF$. Hence, $ED = (EF)(BA)/AF = 18$.

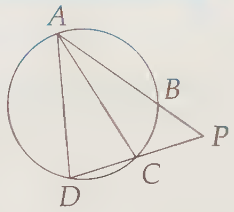
□

We used Power of a Point in combination with similar triangles in the previous problem. Now let's try using it with some of our other geometric tools.

Problem 13.8: In the diagram, we have $BP = 8$, $AB = 10$, $CD = 7$, and $\angle APC = 60^\circ$. Find the area of the circle. *(Source: AHSME)*



Solution for Problem 13.8: It's not immediately obvious how we will find the radius, so we start by finding what we can. The power of a point P gives us



$$(PC)(PD) = (PB)(PA),$$

so $(PC)(PC + 7) = 144$. Therefore, $PC^2 + 7PC - 144 = 0$, so $(PC + 16)(PC - 9) = 0$. PC must be positive, so $PC = 9$.

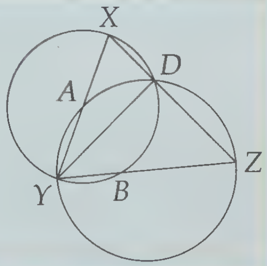
Seeing that $\angle APC = 60^\circ$ makes us wonder if there are any equilateral or 30-60-90 triangles lurking about. Since $CP = AP/2$ and the angle between these sides is 60°, the sides adjacent to the 60° angle in $\triangle ACP$ are in the same ratio as the sides adjacent to the 60° angle in a 30-60-90 triangle. Therefore, $\triangle APC$ is similar to a 30-60-90 triangle by SAS Similarity, so $\triangle APC$ must be a 30-60-90 triangle with right angle at $\angle ACP$!

Since $\angle ACD$ is right and inscribed in $\widehat{AD}$, we know $\widehat{AD}$ is a semicircle. Therefore, $\overline{AD}$ is a diameter of the circle. Since $AC = CP\sqrt{3} = 9\sqrt{3}$ from our 30-60-90 triangle, we have $AD = \sqrt{AC^2 + CD^2} = \sqrt{243 + 49} = 2\sqrt{73}$. Finally, the radius of the circle is $AD/2 = \sqrt{73}$, so the area is $(\sqrt{73})^2\pi = 73\pi$. □

Concept: If you see a 60° or 30° angle (or even a 120° angle) in a problem, you should be on the lookout for 30-60-90 triangles.

Now we will use Power of a Point in a couple proofs.

Problem 13.9: Given that $\overline{YD}$ bisects $\angle XYZ$ in the diagram at right, prove that $XA = ZB$.



Solution for Problem 13.9: This problem involves lengths, an angle bisector, chords, secants, and circles. Therefore, we should consider the Angle Bisector Theorem and Power of a Point. The Angle Bisector Theorem applied to angle bisector $\overline{YD}$ of $\triangle XYZ$ gives us

$$\frac{XD}{XY} = \frac{ZD}{ZY}.$$

However, we need something with XA and ZB. Therefore, we turn to the powers of points X and Z, which respectively give us

$$\begin{aligned}(XA)(XY) &= (XD)(XZ)\\(ZD)(ZX) &= (ZB)(ZY).\end{aligned}$$

Seeing some common terms in these two, along with terms from our Angle Bisector Theorem equation, we know we're on the right track. We solve these two equations for XA and ZB, and we have

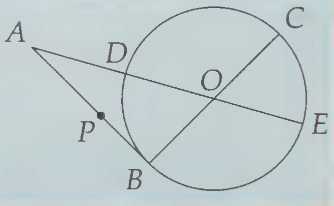
$$\begin{aligned}XA &= \frac{(XD)(XZ)}{XY}\\ZB &= \frac{(ZD)(XZ)}{ZY}\end{aligned}$$

We need only show these two expressions are equal. We have XZ in common, and we already have $XD/XY = ZD/ZY$ from our Angle Bisector Theorem. We can put these together to show

$$XA = \frac{(XD)(XZ)}{XY} = XZ\left(\frac{XD}{XY}\right) = XZ\left(\frac{ZD}{ZY}\right) = \frac{(ZD)(XZ)}{ZY} = ZB.$$

□

Problem 13.10: Point O is the center of the circle, $\overline{AB} \perp \overline{BC}$, $AP = AD$, and $\overline{AB}$ has length twice the radius of the circle. Prove that $AP^2 = (PB)(AB)$. *(Source: AHSME)*



Solution for Problem 13.10: We have a lot of information here. What we want to prove looks a lot like Power of a Point when we have a tangent. $\overline{AB}$ sure looks tangent, but we have to prove it. Fortunately, since diameter $\overline{BC}$ is perpendicular to $\overline{AB}$ at point B on the circle, $\overline{AB}$ must be tangent to the circle. Therefore, $AB^2 = (AD)(AE)$.

This problem has a lot of different segments. We could get lost in a blizzard of AP's and AD's and DE's and so on. So, we have to stay organized. One way to do this is making a list of what we know and what we want to show. Letting the radius of the circle be r, such a table might look like this:

What We Know	What We Want
$AP = AD$ $AB^2 = (AD)(AE)$ $AB = 2r$	$AP^2 = (PB)(AB)$

This has the advantage of letting us work in two directions. We can work forwards from what we know to try to reach what we want, or we can work backwards from what we want. Let's try a little of both here. First, going forwards, we note that $BC = DE = 2r$ because they are diameters of the circle. Combining this with $AB = 2r$ gives $AB = BC = DE$.

Going backwards, we write our desired relationship in terms of fewer lengths. Specifically, since $PB = AB - AP$, we can write $AP^2 = (PB)(AB)$ in terms of just AB and AP. We can also note that $AB = 2r$, and write the equation we want in terms of r and AP. Now our table looks like:

What We Know	What We Want
$AP = AD$ $AB^2 = (AD)(AE)$ $AB = 2r = DE = BC$	$AP^2 = (PB)(AB)$ $AP^2 = (AB - AP)(AB)$ $AP^2 = (2r - AP)(2r)$

Going forwards with our new information, we don't see anything we can do with BC, but $\overline{DE}$ is part of $\overline{AE}$, so we have $AB^2 = (AD)(AE) = (AD)(AD + DE) = (AD)(AD + AB)$. All the stuff on our 'What We Want' side has AP's in it, so we use $AD = AP$ to write

$$AB^2 = (AD)(AD + AB) = (AP)(AP + AB) = AP^2 + (AP)(AB).$$

Now our table looks like this:

What We Know	What We Want
$AP = AD$ $AB^2 = (AD)(AE)$ $AB = 2r = DE = BC$ $AB^2 = AP^2 + (AP)(AB)$	$AP^2 = (PB)(AB)$ $AP^2 = (AB - AP)(AB)$ $AP^2 = (2r - AP)(2r)$

Now we have an equation in our 'What We Know' that is *very* close to an equation in our 'What We Want.' A little rearranging turns one into the other. Subtracting $(AP)(AB)$ from both sides of

$$AB^2 = AP^2 + (AP)(AB)$$

gives $AB^2 - (AP)(AB) = AP^2$. Factoring then gives

$$(AB - AP)(AB) = AP^2,$$

which is the second equation in our 'What We Want' column.

Ah-ha! We have a path from something We Know to something We Want! However, we can't just say 'Done!' We have to write a nice solution. We do so by retracing our steps. We have to examine our work above to figure out three things:

(a) How we get from the given information to the equation $AB^2 = AP^2 + (AP)(AB)$.

(b) How we get from this equation to the one in our list of 'What We Want': $AP^2 = (AB - AP)(AB)$.

(c) How we get from $AP^2 = (AB - AP)(AB)$ to what we want to prove: $AP^2 = (PB)(AB)$.

After reviewing our work above, we write our solution:

Since diameter $\overline{BC}$ is perpendicular to $\overline{AB}$ at point B on the circle, $\overline{AB}$ must be tangent to the circle. Therefore, the power of point A gives us $AB^2 = (AD)(AE)$. We are given AB equals twice the radius of the circle of which $\overline{DE}$ is a diameter, so $AB = DE$. Since we are also given $AD = AP$, we have

$$AB^2 = (AD)(AE) = (AD)(AD + DE) = (AP)(AP + DE) = (AP)(AP + AB).$$

Expanding and rearranging $AB^2 = (AP)(AP + AB)$ gives $AB^2 - (AP)(AB) = AP^2$. Factoring yields $(AB - AP)(AB) = AP^2$. Since $AB - AP = PB$, we have the desired $(PB)(AB) = AP^2$.

Notice that we have some items on our 'What We Know' and 'What We Want' lists that we don't use. This will usually be the case for challenging geometry proofs. □

Important: When writing a solution, you should write a clean solution like what we have after 'After reviewing our work above, we write our solution.' All the work we did before that point was just our investigation to find the solution. This often involves working both forwards and backwards, as we saw. However, once we find our way, we write a nice clean solution *forwards*, starting from what we are given and ending at what we want to prove.

Concept: Organize your work on challenging geometry proofs. Keep track of what you know and what you can derive from what you know. Also keep track of what you need to prove, and work backwards from there to list more statements that, if proven, would mean you are finished. When you know how to get from a statement on your 'What We Know' list to a statement on your 'What We Want' list, then you can construct your proof.

Exercises

13.2.1 In the figure at left below, $\overline{AB}$ is tangent to $\odot O$. Given that $AB = 4$ and $BD = 12$, find the area of the circle.

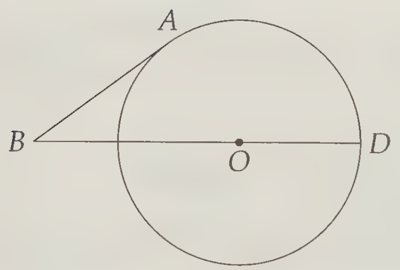


Figure 13.1: Diagram for Problem 13.2.1

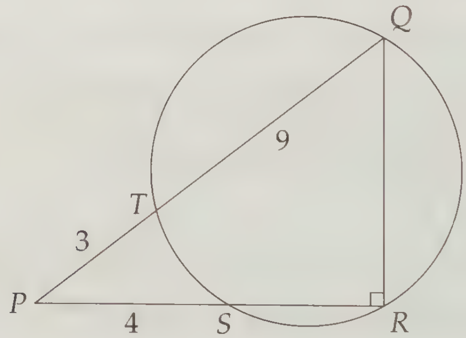


Figure 13.2: Diagram for Problem 13.2.2

13.2.2 Find the following in the figure at right above:

(a) $[PQR]$.

(b) ST. **Hints:** 69

(c) The area of the circle.

(d) $[QTR]$. **Hints:** 594

13.2.3 As shown in the diagram at left below, XA is tangent to the circle at A and a secant through X meets the circle at B and C, with $XB < XC$. Prove that $XB < XA < XC$. **Hints:** 91

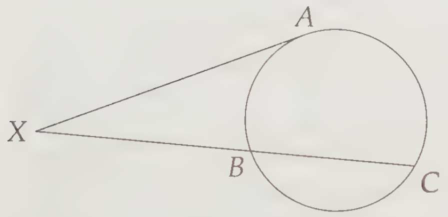


Figure 13.3: Diagram for Problem 13.2.3

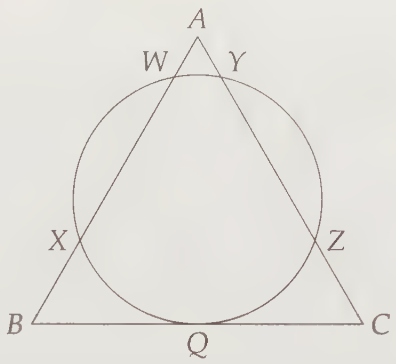


Figure 13.4: Diagram for Problem 13.2.4

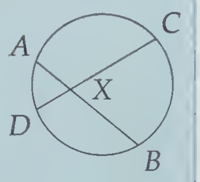
13.2.4 A circle is tangent to side $\overline{BC}$ of equilateral triangle $\triangle ABC$ at point Q as shown at right above. The circle intersects sides $\overline{AB}$ and $\overline{AC}$ in two points each, as shown. Given that $AW = AY$, prove that Q is the midpoint of $\overline{BC}$. **Hints:** 103

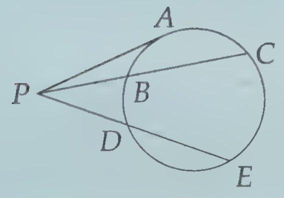
13.3 Summary

Important: Suppose a line through a point P intersects a circle in two points, U and V. The **Power of a Point Theorem** states that for all such lines, the product $(PU)(PV)$ is constant. We call this the **power** of point P.

For example, in the figure at right, applying Power of a Point to X with respect to the circle shown gives

$$(XA)(XB) = (XC)(XD).$$





In the figure at left, the power of point P with respect to the circle gives us

$$PA^2 = (PB)(PC) = (PD)(PE).$$

We think of these whenever we have a problem involving intersecting segments and a circle.

Important: While finding a solution often involves working both forwards and backwards, when we write our solution, we write a nice clean solution *forwards*, starting from what we are given and ending at what we want to prove.

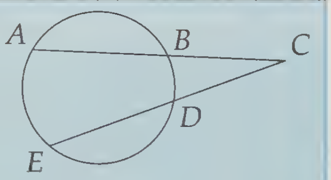
Problem Solving Strategies

Concepts:

- If a problem includes a segment that abruptly stops in the middle of a circle, consider continuing the segment until it hits the circle. Then see what Power of a Point gives you.
- If you see a 60° or 30° angle (or even a 120° angle) in a problem, you should be on the lookout for 30-60-90 triangles.
- Organize your work on challenging geometry proofs. Keep track of what you know and what you can derive from what you know. Also keep track of what you need to prove, and work backwards from there to list more statements that, if proven, would mean you are finished. When you know how to get from a statement on your 'What We Know' list to a statement on your 'What We Want' list, then you can construct your proof.

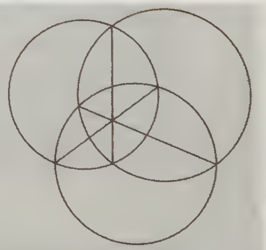
Things To Watch Out For!

WARNING!! A very common mistake in applying Power of a Point to a point outside a circle as in the figure at right is to write $(BC)(BA) = (DC)(DE)$. This alleged equality is **not** what Power of a Point tells us! Note that whenever we write a Power of a Point relationship, the same point must appear in *all* of the segments in the equation, as point C does when we write the correct relationship for the secants in the figure above:



$$(CD)(CE) = (CB)(CA).$$

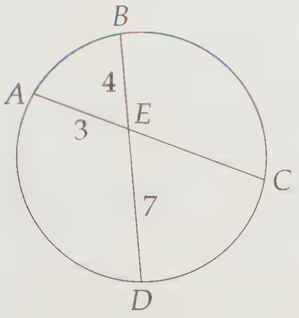
Extra! The set of all points that have the same power with respect to two given circles is a line called the **radical axis** of the two circles. One particularly important use of radical axes is the **Radical Axis Theorem**, which states that given three circles, the three radical axes of the three pairs of circles are concurrent. The diagram at right is an example of the Radical Axis Theorem for three intersecting circles. Do you see why the radical axis of two intersecting circles must include the chord connecting the points where the circles meet?



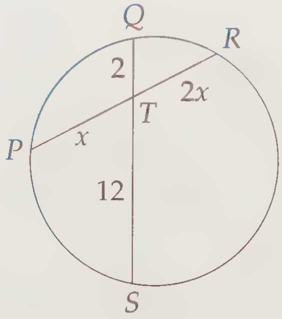
As an extra challenge, try proving first that the set of all points that have the same power with respect to two given circles is a line, then use this to prove the Radical Axis Theorem. (The first part is tougher than the second!)

REVIEW PROBLEMS

13.11

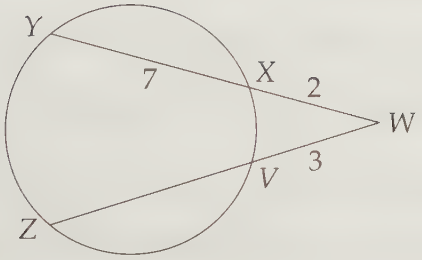


(a) Find EC.

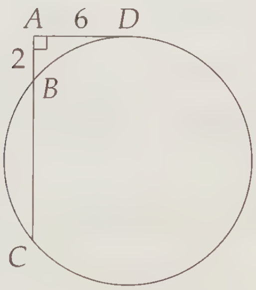


(b) Find PR.

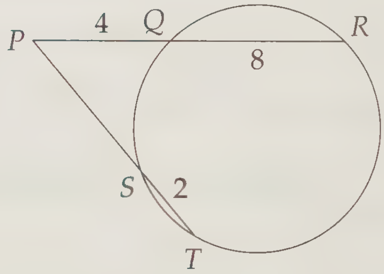
13.12



(a) Find VZ



(b) Find BC and CD.



(c) Find PS.

13.13 Chords $\overline{GH}$ and $\overline{IJ}$ of $\odot O$ are perpendicular at M. Given $GM = 2MH = 6IM = 12$, find GJ. **Hints:** 76

13.14 Lines m and n meet at P. $\odot O$ meets m at points A and B, and meets n at C and D, such that $PA = 3$, $AB = 9$, $PC = 4$, and $CD = 5$. Why must P be outside $\odot O$?

13.15 Earth is roughly 8000 miles in diameter.

(a) I'm riding in a hot air balloon 1 mile above the surface of Earth. Approximately how far away is the horizon? (In other words, how far away is the farthest point on the surface of Earth that I can see.)

(b) What if I'm in a plane 6 miles above the surface of the Earth?

(c) What if I'm in a spaceship 100 miles above the surface of Earth?

13.16 Given $ZY < ZW$ in the diagram at left below, show that $ZV < ZX$.

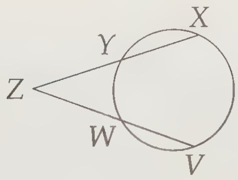


Figure 13.5: Diagram for Problem 13.16

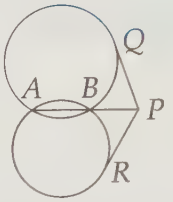
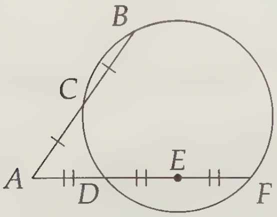


Figure 13.6: Diagram for Problem 13.17

13.17 The figure at right above shows two circles that intersect at A and B. P is on $\overleftrightarrow{AB}$, and $\overline{PQ}$ and $\overline{PR}$ are tangents as shown. Prove that $PQ = PR$.

13.18 Is it possible for chords $\overline{AB}$ and $\overline{CD}$ of a circle to intersect at X such that X is the midpoint of $\overline{AB}$, but is closer to C than to D?

13.19 Is it possible for two chords $\overline{AB}$ and $\overline{CD}$ of a circle to meet at a point X such that $AX = BX$, $CX = 2DX$, and $AB = CD$?

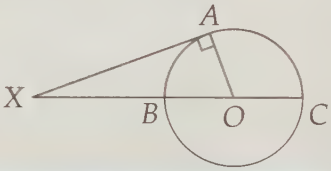


13.20 Find the ratio AC/AD in the figure at left.

13.21 Point A is on $\odot O$ such that $\overline{PA}$ is tangent to circle $\odot O$. Point B is on circle $\odot O$ such that $PB = PA$. Must $\overline{PB}$ be tangent to $\odot O$?

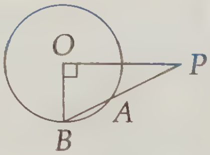
13.22 Let P be a point in the same plane as a circle centered at O with radius r. A secant through P intersects the circle at points A and B. Use $\overleftrightarrow{PO}$ to prove that if P lies outside the circle, then $PA \cdot PB = PO^2 - r^2$. What is the formula if P lies inside the circle?

13.23 Use the diagram in the figure at right to find a proof of the Pythagorean Theorem.



Challenge Problems

13.24 In circle O, $\overline{PO} \perp \overline{OB}$ and PO equals the length of the diameter of $\odot O$. Compute PA/AB. *(Source: ARML)* **Hints:** 294, 323



13.25 Chords $\overline{PQ}$ and $\overline{RS}$ of a circle meet at point X. Given that $PQ = RS$, show that $PX = RX$ or $QX = RX$ (or both). **Hints:** 364

13.26 Jake is working on a problem in which chords $\overline{AB}$ and $\overline{CD}$, when extended past B and D, respectively, meet at P. He is given PB, AB, and CD. He mistakenly uses Power of a Point improperly by using $(PB)(BA) = (PD)(DC)$. However, he still gets the right answer for PD. Prove that the question Jake was answering must have had $AB = CD$. **Hints:** 385

13.27 A and B are two points on a circle with center O, and C lies outside the circle, on ray $\overrightarrow{AB}$. Given that $AB = 24$, $BC = 28$, and $OA = 15$, find OC. **Hints:** 122

13.28 Circles C_1 and C_2 have the same center, O. The radius of C_1 is r_1 and the radius of C_2 is r_2, where $r_1 > r_2$.

(a) Prove that it is impossible for a point P to have the same power with respect to both circles if P is outside both circles. **Hints:** 104

(b) Prove that it is impossible for a point P to have the same power with respect to both circles if P is inside both circles.

(c) Show that it is possible for a point inside C_1 but outside C_2 to have the same power with respect to both circles, and find all such points that have equal power with respect to both circles.

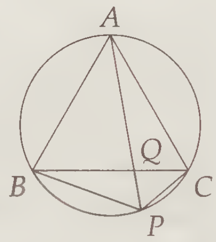
13.29 $\overline{WX}$ and $\overline{YZ}$ meet at P such that $(WP)(PX) = (YP)(PZ)$. Prove that the circumcircle of $\triangle XYZ$ goes through point W. **Hints:** 512

13.30 Two circles C_1 and C_2 intersect at two points, A and B. Let $\overline{PQ}$ be a chord of C_1 and $\overline{RS}$ a chord of C_2 such that they intersect on $\overline{AB}$. Prove that points P, Q, R, and S all lie on a circle. **Hints:** 540, 397

13.31 Given two segments with lengths a and b, construct a segment with length $\sqrt{ab}$. **Hints:** 415

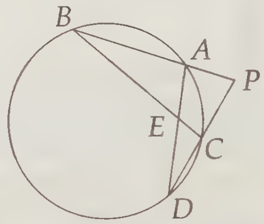
13.32 Equilateral triangle ABC is inscribed in a circle. Let P be a point on arc $\widehat{BC}$, and let $\overline{AP}$ intersect $\overline{BC}$ at Q. Prove that

$$\frac{1}{PQ} = \frac{1}{PB} + \frac{1}{PC}.$$



Hints: 382, 349

13.33 In the diagram at right, $AB = 8$, $AP = 2$, and $PC = 4$. *(Source: ARML)*



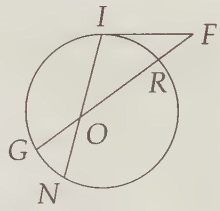
(a) Prove that $[ABE]/[CDE] = (AB/CD)^2$.

(b) Prove that $[PBC]/[PAD] = (PC/PA)^2$.

(c)★ Find $[PAEC]/[BAE]$. **Hints:** 309, 346

13.34★ An equilateral triangle is inscribed in a circle. Points D and E are midpoints of $\overline{AB}$ and $\overline{BC}$, respectively, and F is the point where $\overrightarrow{DE}$ meets the circle. Find DE/EF. *(Source: ARML)* **Hints:** 300, 252

13.35★ Segment $\overline{IF}$ is tangent to the circle at point I as shown in the diagram at right. We are given that $IF = 21\sqrt{2}$, $IO = 20$, $ON = 12$, $RF = 18$, and $OR > GO$. Find OR. *(Source: Mandelbrot)* **Hints:** 188



13.36★ Chords $\overline{AB}$ and $\overline{CD}$ of $\odot O$ are perpendicular at point P. Given $CP = 2$, $AP = 3$, and $PD = 6$, find the following:

(a) OP. **Hints:** 31, 90

(b) The radius of $\odot O$. **Hints:** 387